



EXPLORATORY DATA ANALYSIS

EDA Insights: Vendor & Inventory Performance

10,692 transactions analyzed across vendors, brands, pricing, and profitability.

Summary Statistics: Key Red Flags

Negative Gross Profit

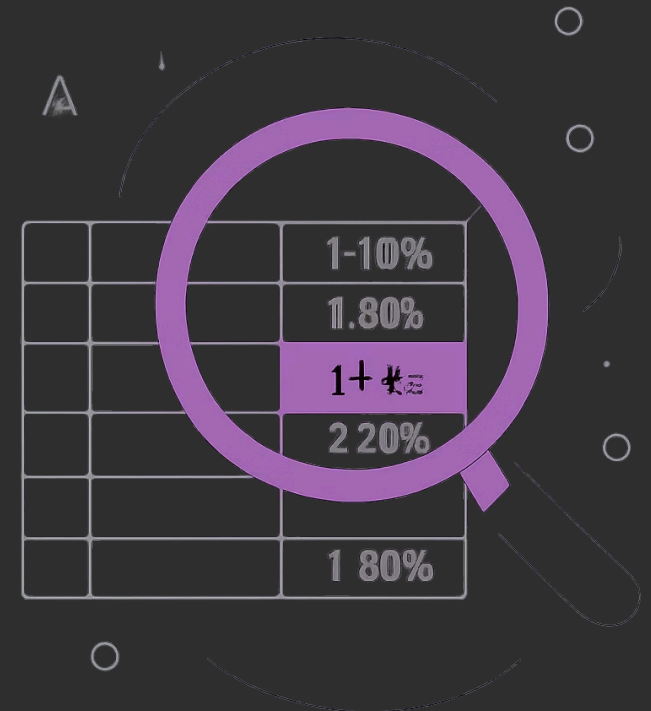
Min of **-\$52,003** — selling below purchase cost.

Extreme Profit Margins

Min of **-2.4M%** — zero-revenue transactions.

Unsold Inventory

Products with **zero sales** — obsolete or slow-moving stock.



Outliers & Extreme Variation

Premium Pricing

Max prices reach \$5,682 vs. mean of \$24.

High standard deviations signal premium product offerings, logistics inefficiencies, and erratic shipping costs across the catalog.

Freight Volatility

Ranges from \$0.09 to \$257,032.

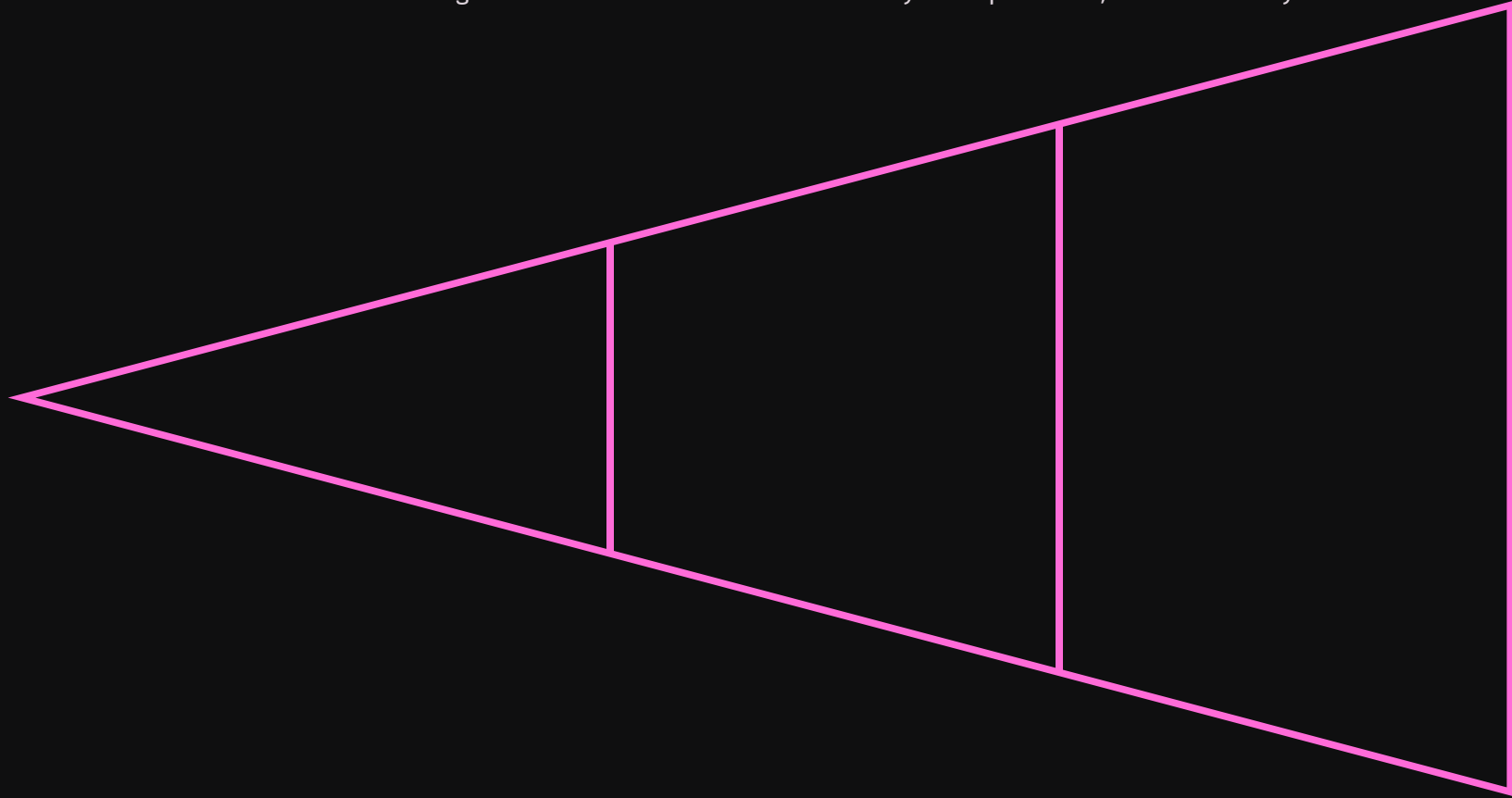
Stock Turnover

0 to 274.5 — rapid sellers vs. dead stock.



Data Filtering for Reliable Insights

Excluded loss-making and unsold transactions to focus analysis on profitable, active inventory.



Correlation Insights

Purchase ↔ Sales Quantity

0.999 correlation — efficient inventory turnover confirmed.

Sales Price ↔ Margin

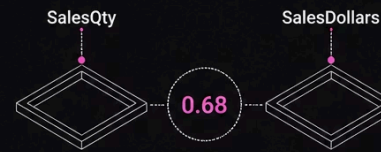
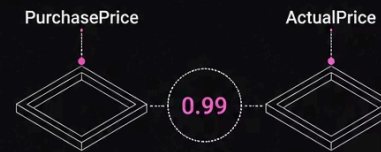
Negative (-0.18) — higher prices may reduce margins.

Price ↔ Sales/Profit

Weak correlation (~-0.01) — price changes barely impact revenue.

Turnover ↔ Profitability

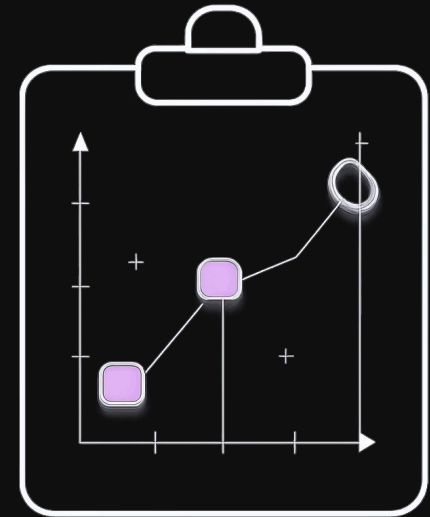
Weak negative — faster turnover ≠ higher profit.



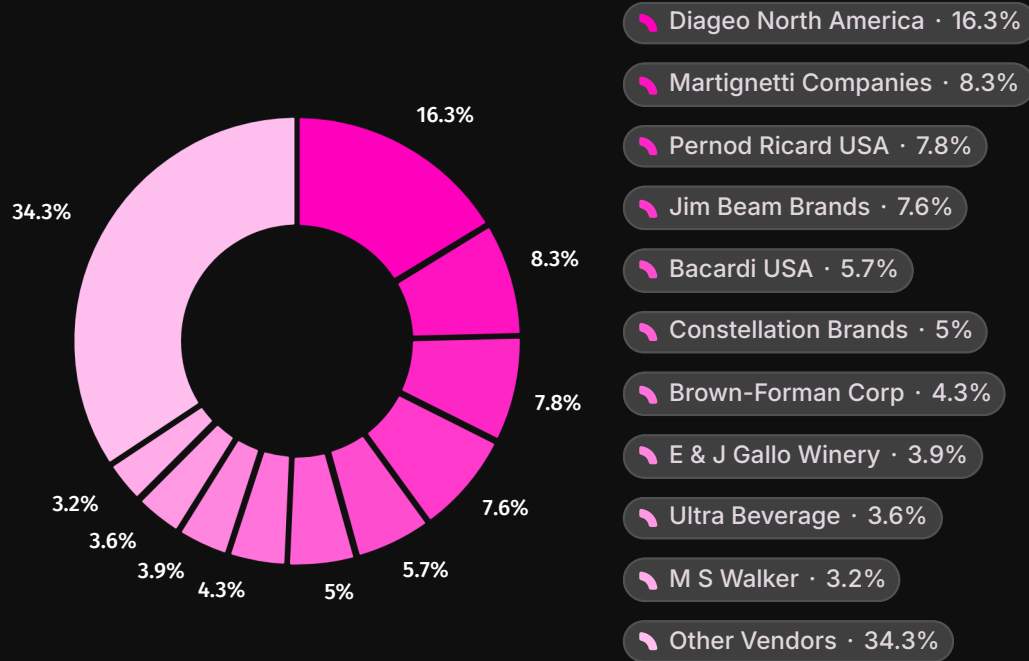
Brands for Promotional Adjustments

198 brands with low sales but high profit margins (above 65.69%).

Clustered in upper-left quadrant — ideal for targeted marketing and price optimization.



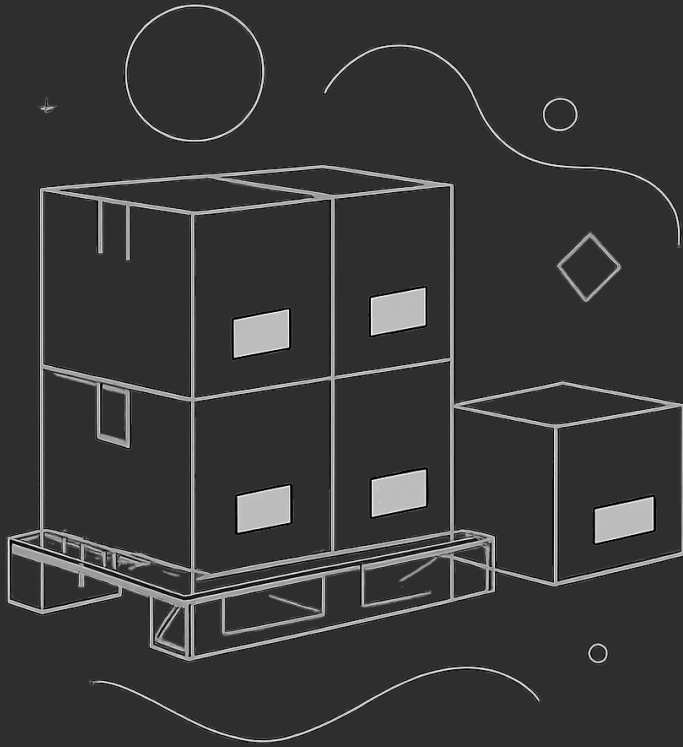
Top 10 Vendors: Purchase Contribution



Top 10 vendors contribute **65.69%** of total purchases.

Over-reliance on few vendors risks supply chain disruptions — diversification needed.

Bulk Purchasing & Inventory Turnover



72%

Lower Unit Cost

Large orders vs. small orders

\$10.78

Bulk Unit Price

vs. \$39.07 for small orders

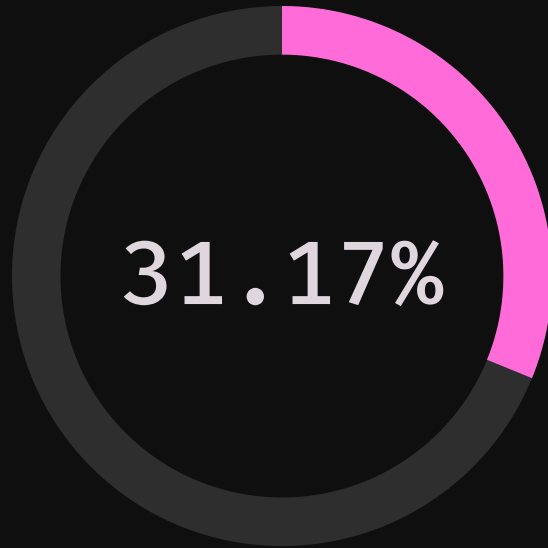
\$2.71M

Unsold Capital

Tied up in slow-moving stock

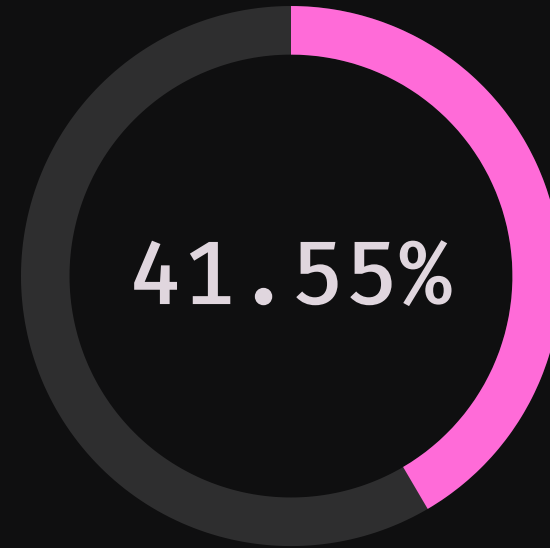
Bulk pricing encourages large orders while maintaining profitability. Low turnover vendors like Alisa Carr Beverages (0.62) and Highland Wine (0.71) tie up capital in unsold inventory.

Profit Margin: Top vs. Low Vendors



Top Vendors Mean

95% CI: 30.74%–31.61%



Low Vendors Mean

95% CI: 40.48%–42.62%

Low-performing vendors maintain **higher margins** but struggle with sales volume — indicating pricing inefficiencies or limited market reach.



Null hypothesis rejected: the two groups operate under distinctly different profitability models.



FINAL RECOMMENDATIONS

Actionable Strategy & Power BI



Optimize Pricing

Reassess low-sales, high-margin brands to boost volume.



Diversify Vendors

Reduce dependency on top 10 to mitigate supply risk.



Bulk & Logistics

Leverage bulk purchasing; control freight cost variations.



Monitor KPIs

Track margins, turnover, and inventory via Power BI dashboards.